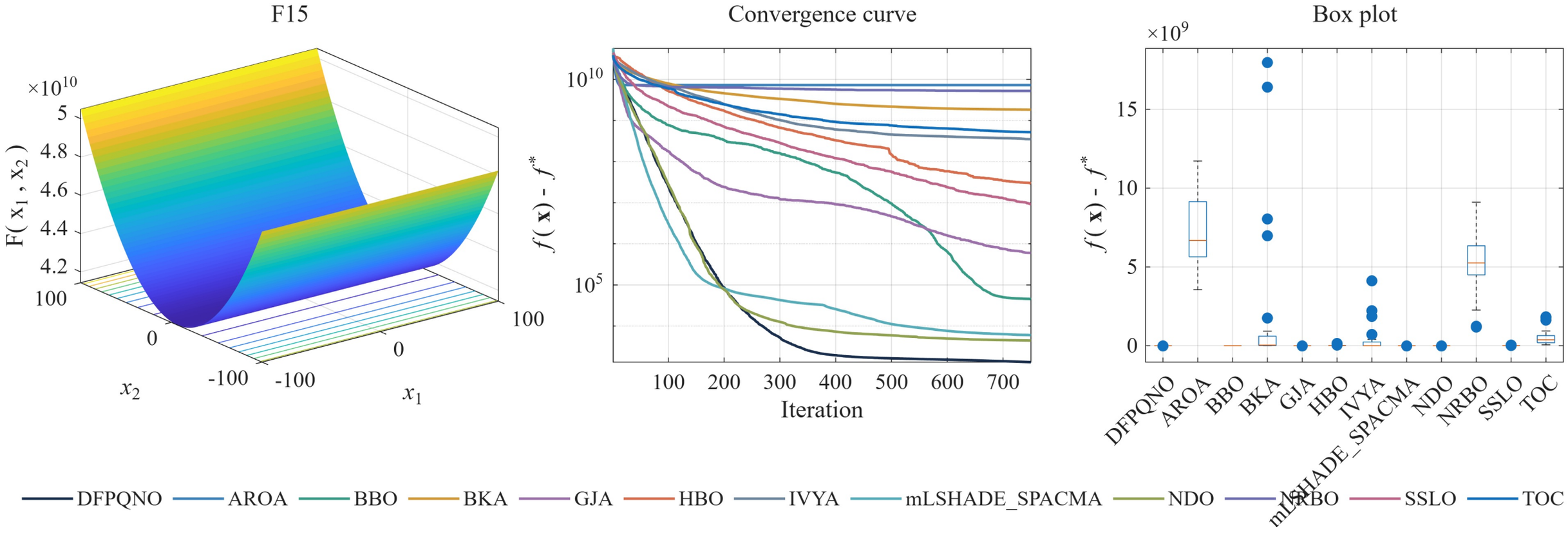

F15
×10^10
5
4.8
4.6
4.4
4.2
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
10^10
10^5
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
×10^9
15
10
5
0
f(x) - f*
DFPQNO
AROA
BBO
BKA
GJA
HBO
IVYA
mLSHADE_SPACMA
NDO
NRBO
SSLO
TOC
DFPQNO
AROA
BBO
BKA
GJA
HBO
IVYA
mLSHADE_SPACMA
NDO
NRBO
SSLO
TOC